

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 6

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Suppose you have 50 worker processes trying to access only 4 available accelerators.
 - A. Using a counting semaphore, write the pseudo-code for a worker process to acquire an accelerator, perform a calculation, and release it.
 - B. If you accidentally initialized the semaphore to 0, what would happen to the 50 workers?
 - C. What happens to the system performance if a worker process crashes after calling P() but before calling V()?
2. We saw that `counter++` is not atomic. Suppose you have two processes, Process A and Process B. Both execute `counter++` 1,000,000 times.
 - A. Place a binary semaphore (mutex) to make the increment safe.
 - B. Your friend suggests moving the P() call outside the loop (before the million increments start) and V() after the loop ends. Does this still produce the correct result of 2,000,000?
3. Processes P1, P2, and P3 are part of a data pipeline with the following rules:
 - P1 must print "Ready".
 - P2 can only print "Set" after P1 finishes.
 - P3 can only print "Go!" after P2 finishes.
 - A. Initialize the necessary semaphores and add the P and V calls to each process to enforce this exact order.
 - B. Modify your solution so that P3 can start only after both P1 and P2 have finished their work, but P1 and P2 are allowed to run at the same time.
4. In the Readers/Writers Problem with Reader Priority, multiple readers can access the database at once.
 - A. Describe a specific sequence of events where a Writer process might wait for an infinite amount of time (Starvation).
 - B. If you were designing an airline reservation system where price updates (Writes) are just as important as searches (Reads), would you use Reader Priority or Writer Priority? Justify your choice based on the logic in Slide 27 of Lecture 6.
5. Implement a synchronized data relay that uses POSIX semaphores to coordinate a Producer-Consumer relationship between two processes. The goal is to enforce a strict sequence where the producer process generates a random number into a shared variable only when the buffer is empty, while the consumer process waits for data to arrive and process it. The output must show a strict alternation between the

producer and consumer, proving that no data is overwritten or read twice. Each process must print its action to stdout to demonstrate synchronization.

- **Welcome: Sean**

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